

Modeling NBA Player Performance Using Structural Equation Modeling

Computer Project - SEM Course May 2026

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Abstract

Basketball is today one of the biggest sports on the planet and has a viewership all over the globe. Basketball is played, as for many other sports, by attacking the opponents goal and scoring point, whilst trying to defend ones own goal. This report uses structural equation modeling (SEM) and confirmatory factor analysis (CFA) to try to identify underlying dimensions in NBA player performances based on game statistics from the 24/25 season. The report specifically investigates whether player statistics can be represented by latent dimensions related to abilities offensively and defensively on the court. It further investigates whether the measurement structure differs between rookie and veteran players and whether one of the latent constructs has any predictive power of the other. The results suggest that basketball is not defined by offensive and defensive ability, but rather by player archetypes. It further demonstrates that trying to compare a small group of rookie players with a much larger group of veterans is incredibly problematic when looking for significant results. It finally shows that the revised offensive factor, which focuses more on involvement in play, has a significant predictive power of the revised defensive factor, which itself focuses much more on positioning on the field rather than pure defensive ability. By applying CFA, SEM, and multigroup analysis, the study demonstrates how the framework can be used to model complex performance structures in sports analytics.

1 Introduction

Basketball performance is widely regarded as multidimensional, with players contributing through different combinations of scoring, playmaking, rebounding, and defensive actions. Because individual statistics capture only specific aspects of performance, basketball analytics often focuses on relationships among multiple indicators rather than single metrics [Oliver, 2004]. This motivates the use of latent variable models to identify broader performance dimensions underlying observed NBA statistics. Studying such dimensions might help explaining what a good basketball player is defined by. By evaluating an initial offense-defense structure and examining whether the measurement structure differs between rookies and veterans this study aims to explore this concept. The main objective of the study is thus to answer the following research questions:

- Can basketball statistics be explained by offensive and defensive ability of a player?
- Can an offensive latent construct be used as predictor for a defensive latent construct among NBA players?
- Is there a statistically significant difference in latent measurement structure between rookies and veteran players?

2 Data

The data used in this project consist of NBA player game statistics from the 2024–25 season. Each row in the original dataset represents a player performance in a specific game, meaning that the data are organized at the player-game level (Palmieri, 2025). The dataset includes variables such as player name, minutes played, shooting statistics, rebounds, etc., and is relevant to the research questions as it contains observed basketball statistics that can be used as indicators of broader latent player abilities. For example, points and assists could be viewed as indicators of offensive ability, whilst rebounds and steals could indicate defensive ability.

The original dataset contains 28,265 player-game observations and 27 variables. Since the analysis focuses on overall player ability rather than single-game performance, the data are aggregated to the player-level. For each player, the selected statistics are averaged across all games in which the player appeared. This results in a final player-level dataset containing 569 unique players. The variables used in the CFA model are points (PTS), field goal percentage (FG%), three-point percentage (3P%), assists (AST), offensive rebounds (ORB), free throws (FT), steals (STL), blocks (BLK), defensive rebounds (DRB), total rebounds (TRB), and turnovers (TOV).

Table 1: Overview of Original Dataset From the 24/25 NBA Season

Item	Description
Original sample size	28,265 observations
Number of variables	27
Main variables used	PTS, FG%, 3P%, AST, ORB, FT, STL, BLK, DRB, TRB, TOV

Missing data are identified by checking the number of missing values in the selected variables. Only variables to contain missing values are FG% and 3P%. This is expected in basketball data because a shooting percentage can be undefined when a player has no attempts in the category. When calculating player-level averages, missing values are handled using `na.rm = TRUE` in R, meaning that all available observations are used for each player.

There are some limitations in the data. First, box-score statistics do not capture every aspect of basketball performance, such as defensive positioning, off-ball movement, screening, communication, or tactical role. Second, the number of games played differs across players, so players with fewer games may have less stable average statistics. Third, percentage statistics can be affected by low shot volume, meaning that a player with few attempts may have an extreme shooting percentage that does not necessarily reflect long-term ability. Finally, The original dataset does not include a direct variable for player experience. Thus, a rookie/veteran grouping-variable is created manually where players who were first-year NBA players in the 24/25 season are classified as rookies, while all remaining players are classified as veterans. Rookie status is coded as a binary grouping variable in R, where rookies are coded as 1 and veterans as 0. In total, the data set is divided into 20 rookies and 549 veterans.

Table 2: Descriptive Statistics of Selected Player-Level Variables

	PTS	FG%	3P%	AST	FT	TOV	ORB	DRB	TRB	STL	BLK
<i>n</i>	569	560	541	569	569	569	569	569	569	569	569
Mean	8.800	0.437	0.301	2.050	1.320	1.080	0.930	2.640	3.570	0.660	0.397
Median	7.170	0.430	0.328	1.490	0.867	0.871	0.724	2.360	3.120	0.621	0.280
SD	6.560	0.115	0.121	1.800	1.350	0.830	0.775	1.800	2.420	0.420	0.402
Kurtosis	0.559	3.640	4.780	2.880	3.880	1.460	2.980	2.050	1.800	1.490	11.700
SE	0.275	0.005	0.005	0.076	0.057	0.035	0.032	0.075	0.101	0.018	0.017

Table 2 summarizes the player-level variables used in the analysis. Several variables exhibit substantial kurtosis, motivating the use of robust MLR estimation in the rest of the report.

3 Theory

3.1 SEM and CFA Theory

The multivariate statistical framework implies that observed values might be indicators of underlying theoretical constructs which can be modeled. To do so, confirmatory factor analysis (CFA) is used. Performing the analysis, the factor structure is theoretically predefined and then empirically tested (Jöreskog et al., 2016). The factor model in this report can be expressed as:

$$x = \Lambda\xi + \delta$$

where x represents player statistics, Λ represents the factor loading matrix, ξ represents basketball ability and δ is the measurement error. CFA further evaluates how well the model-implied covariance matrix reproduces the observed covariance matrix. Jöreskog et al. (2016). The model-implied covariance matrix can be expressed as:

$$\sum(\theta) = \Lambda\Phi\Lambda^T + \Theta_\delta$$

$\sum(\theta)$ represents the model-implied covariance matrix, whilst Λ is the factor loading matrix and Φ is the covariance matrix of the latent variables. Θ_δ finally represents the measured error variances in the model.

After establishing a measurement model, a structural regression path can be added from one latent variable to another. The regression part of the model can be expressed as:

$$Factor1 = \beta(Factor2) + \zeta$$

where β represents the latent regression coefficient and ζ represents residual variance in factor 1 not explained by factor 2.

A potential issue in CFA and SEM is the occurrence of Heywood cases, where estimated variances become negative or other inadmissible parameter estimates arise. Such results may indicate model misspecification or estimation problems and should therefore be interpreted with caution (Farooq, 2022).

3.1.1 Multi-Group CFA

Multi-group CFA is used to examine whether latent constructs behave similarly between specific groups. It is based on measurement invariance theory, which evaluates whether latent constructs are measured equivalently across different populations (Jöreskog et al.,

2016). Measurement invariance is necessary because meaningful comparisons between groups require that the same latent constructs are represented by the same observed variables across populations. Because of this, this analysis follows the standard measurement invariance procedure by testing configural, metric, and scalar invariance. These models assess whether factor structure, loadings, and intercepts are equivalent across rookie and veteran players Jöreskog et al. (2016).

3.2 Model Identification

For the model to be identified, the number of unique elements in the observed covariance matrix must exceed the number of free parameters estimated:

$$\frac{p(p+1)}{2} > q$$

p denotes the number of observed variables and q the number of free parameters. To identify the latent variables, one factor loading per latent construct is fixed to 1, thereby defining the measurement scale of each factor through a reference indicator.

3.3 Estimation Technique

The CFA model is estimated using robust maximum likelihood estimation (MLR) implemented in the lavaan package in R. Robust MLR is implemented as it accounts for deviations from multivariate normality in the data. Maximum likelihood estimation further provides stable, consistent estimates. The estimation is performed by minimizing the fit function (Jöreskog et al., 2016):

$$F_{ML} = \log ||\Sigma|| + \text{tr}(S\Sigma^{-1}) - \log ||S|| - p$$

where S is the observed covariance matrix and Σ is the model-implied covariance matrix. Full information maximum likelihood (FIML) is used to deal with missing observations. FIML estimates model parameters directly from all available data without deleting incomplete cases (Jöreskog et al., 2016).

3.4 Evaluation Measurements

Cronbach's alpha is used before the CFA is conducted, and is a commonly used measure of internal consistency that evaluates the extent to which a set of observed indicators measures the same underlying construct (Cronbach, 1951). The evaluation measures assess the fit of

the model by evaluating how closely the covariance matrix implemented in the model reproduces the observed covariance matrix (Jöreskog et al., 2016). The chi-square test statistic evaluates exact model fit:

$$\chi^2 = (N - 1)F_{ML}$$

N denotes sample size. Because χ^2 is sensitive to sample size, additional fit indices are evaluated. The following are used:

$$CFI = 1 - \frac{\max(\chi_m^2 - df_m, 0)}{\max(\chi_m^2 - df_m, \chi_b^2 - df_b, 0)}$$

Comparative Fit Index

$$TLI = \frac{\chi_m^2/df_m - 1}{\chi_b^2/df_b - 1}$$

Tucker-Lewis Index

$$RMSEA = \sqrt{\max\left(\frac{\chi_m^2 - df_m}{df_m(n - 1)}, 0\right)}$$

Root Mean Square Error of Approximation

$$SRMR = \sqrt{\frac{2}{p(p + 1)} \sum_{i < j} \frac{(s_{ij} - \hat{\sigma}_{ij})^2}{s_{ii}s_{jj}}}$$

Standardized Root Mean Square Residual

For the comparative fit index (CFI) and the Tucker-Lewis Index (TLI), m denotes the hypothesized model and b is the baseline model. Approximate fit is evaluated using RMSEA and the Standardized Root Mean Square Residual (SRMR) is also used, where s_{ij} represents the observed covariances and $\hat{\sigma}_{ij}$ are the model-implied covariances. Interpretation as well as satisfactory values for each measurement as discussed by Hu and Bentler (1999) are presented in table 3.

Table 3: Goodness of Fit Criteria for CFA Model

Fit index	Good fit	Acceptable fit	Interpretation
CFI	$\geq .95$	$\geq .90$	Comparative fit
TLI	$\geq .95$	$\geq .90$	Incremental fit
RMSEA	$\leq .05$	$\leq .08$	Approximate fit
SRMR	$\leq .05$	$\leq .08$	Residual fit
χ^2	Non-significant	—	Exact fit

4 Results

4.1 Initial Model

An original two-factor model is firstly proposed for the latent structure in basketball ability. The proposed structure follows from understandings of how the game is played. In the

proposed model, two latent variables (offensive- and defensive ability) are theorized to be larger constructs for observed player performances. For offensive ability, the suggested model implies that the variables PTS, AST, FT, FG% and 3P% load onto the factor. Variables STL, BLK, DRB and TRB are suggested to load onto defensive ability. To examine the theorized structure, a correlation matrix of the variables is examined.

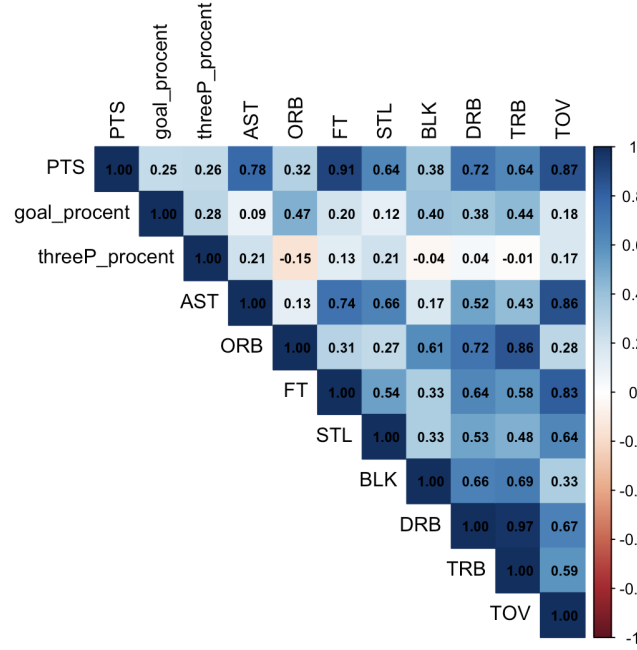


Figure 1: Correlation Matrix

Figure 1 shows that the correlation structure in the data fits relatively well with the hypothesized model. Large correlations are observed between most defensive variables, with STL and BLK correlating the lowest at 0.33. Initial refinement is conducted to account for the low correlation of FG% and 3P% with all other variables under the same latent factor. The two variables are removed from the initial model as low correlation suggests a bad fit under this model construction. In contrast, DRB and total rebounds TRB exhibits an extremely high correlation with 0.97, indicating overlap between the variables. Although DRB and TRB exhibits extremely high correlations, a residual covariance is not specified as the relationship reflects overlap rather than covariance between the two indicators. A correlated residual is however specified between PTS and FT due to their relationship in the game. Cronbach's alpha is calculated as a measure of internal consistency (Cronbach, 1951) prior to the CFA analysis.

Table 4: Cronbach’s Alpha for Proposed Latent Constructs

Model	Construct	Standardized α
Initial model	Offense	0.93
Initial model	Defense	0.86

Both proposed constructs demonstrates high internal consistency (table 4), however, final evaluation of the latent constructs is based on CFA model fit and factor loadings.

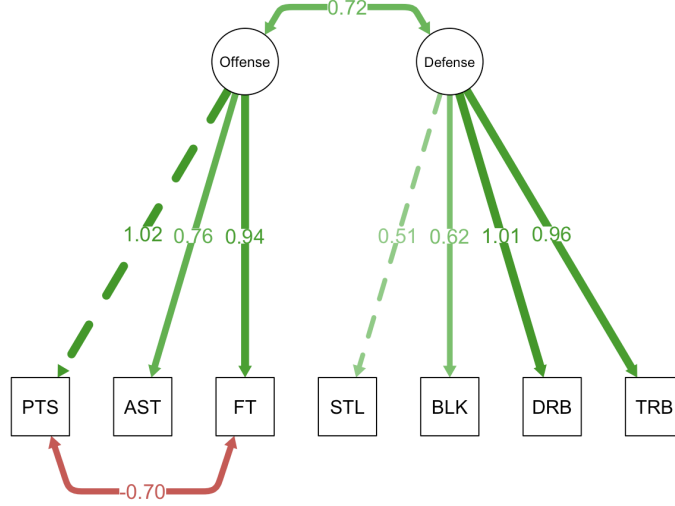


Figure 2: Initial Proposed CFA model

Table 5: Evaluation Measurements for Initial CFA Model

	χ^2	df	p	CFI	TLI	RMSEA	SRMR
Standard	440.661	12	< .001	0.907	0.837	0.251	0.096
Robust	247.306	12	< .001	0.908	0.839	0.249	0.096

The model satisfies the t-rule, since the number of observed variances and covariances exceeded the number of free parameters and is thus identified. Despite the high reliability estimates, the initial CFA model shows inadequate fit overall. Although the CFI is close to the commonly used threshold for acceptable fit, the TLI is below acceptable levels and both RMSEA and SRMR indicates substantial model misspecification. The high value of the robust RMSEA (0.249), suggests that the hypothesized offense-defense structure does not adequately reproduce the observed covariance matrix. In addition, heywood cases are produced in the model, with negative residual variances for both PTS and DRB. This indicates instability in the model (Farooq, 2022), and suggests that the initial theoretical distinction between offensive and defensive ability not fully captures the structure of NBA statistics. Several standardized loadings moreover approaches or exceeded unity, further indicating

instability in the latent structure, supporting model refinement.

4.2 Revised CFA Model

Based on the correlation matrix and modification indices the model is revised. The correlation structure suggests that offensive production variables cluster more strongly around player involvement and ball usage rather than offensive efficiency. For defensive variables, a sharper clustering is also suggested around interior play, around the hoop rather than all out defense. Instead of traditional offense-defense separation, the revised model thus specifies two role-based latent constructs: offensive involvement and interior presence. Offensive involvement is now suggested to capture involvement in play rather than efficiency and ability offensively, and is measured by PTS, AST, FT, and TOV. Interior presence is measured by ORB, BLK, and DRB. Residual covariances are added between AST and TOV, PTS and FT, and ORB and BLK. The residual covariance between AST and TOV reflects that players with high playmaking responsibility also tend to commit more turnovers due to increased offensive decision-making. Similarly, the covariance between PTS and FT captures shared scoring-related variance not fully explained by the latent offensive involvement factor. The covariance between ORB and BLK likely reflects shared interior-player characteristics related to positioning and proximity to the basket (Oliver, 2004).

Table 6: Cronbach’s alpha for revised latent constructs

Model	Construct	Standardized α
Revised model	Offensive Involvement	0.95
Revised model	Interior Presence	0.86

The revised role-based model demonstrates strong internal consistency, particularly for the construct of offensive involvement ($\alpha = 0.95$). The model is from here changed and fitted as a SEM model.

Table 7: Evaluation Measurements for Revised CFA Model

	χ^2	df	p	CFI	TLI	RMSEA	SRMR
Standard	92.744	10	< .001	0.979	0.956	0.121	0.039
Robust	61.396	10	< .001	0.981	0.960	0.116	0.039

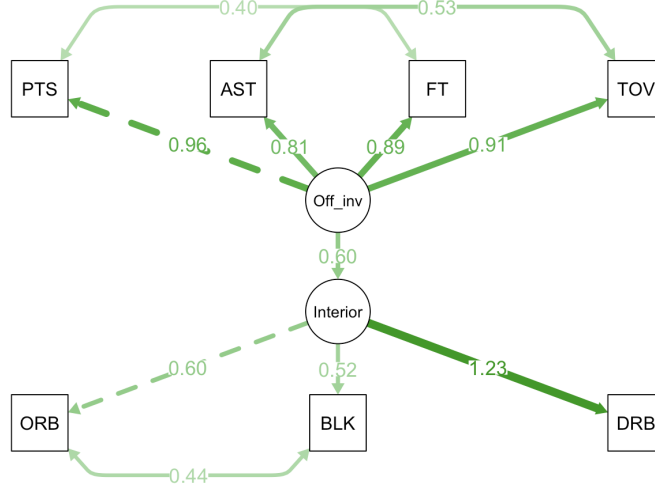


Figure 3: Revised Structural SEM Model

The revised CFA model remains overidentified ($df = 10$). The model shows a substantial improvement in fit compared to the initial model with robust CFI (0.981) and TLI (0.960) indicating good fit, while the SRMR (0.039) suggests that the model reproduces the observed correlation structure well. Although RMSEA remains large at 0.116, the overall pattern of the evaluation measurements supports the revised model as a considerably better representation of the data and is also extended into a structural SEM model by adding a regression path from Offensive Involvement to Interior Presence.

The revised latent structure appears to capture role specialization more effectively than the initial offense-defense distinction. The offensive involvement factor primarily reflects ball-dominant offensive responsibility, including scoring production, playmaking, and turnover generation.

Similarly, the Interior factor captures physical interior presence through rebounding and rim protection. The strong loading of DRB suggests that defensive rebounding dominates this latent construct, reflecting the importance of size and positional interior play in NBA performances.

Table 8: Structural Regression Estimates for the Revised SEM Model

Structural Path	Estimate	SE	z-value	p-value	Std. β
Interior Pre $\leftarrow$ Offensive Inv	0.374	0.042	8.916	< .001	0.603

Table 8 shows that the standardized regression coefficient from Offensive Involvement to Interior Presence is positive and statistically significant, indicating that players with higher offensive involvement also tend to have higher levels of interior presence. This means that highly involved offensive players are not only more active in scoring and playmaking, but also tend to contribute more strongly to interior aspects of play such as rebounding and shot

blocking. However, the result should be interpreted with caution because the model still produced a negative residual variance for defensive rebounds (DRB), indicating a Heywood case. This suggests some instability in the Interior Presence factor, likely due to the very strong relationship between defensive rebounds and defense as a latent construct overall.

4.3 Rookies vs Veterans

Measurement invariance is assessed using configural, metric, scalar, and residual invariance models.

Table 9: Fit Indices for Multi-Group CFA Invariance Models

Model	χ^2	df	CFI	RMSEA	SRMR	TLI
Configural	104.188	20	0.979	0.122	0.043	0.956
Metric	114.645	25	0.978	0.112	0.040	0.962
Scalar	128.210	30	0.975	0.107	0.041	0.966
Residual	145.739	37	0.973	0.102	0.046	0.969

Table 9 shows that all models exhibits high CFI and TLI values, and low SRMR values. However, RMSEA is relatively high across all models, suggesting that model fit is not perfect. This should be considered when interpreting the results.

Table 10: Chi-Square Difference Tests for Measurement Invariance

Comparison	$\Delta\chi^2$	Δdf	p-value	Decision
Configural vs Metric	10.456	5	0.063	Not rejected
Metric vs Scalar	13.566	5	0.019	Rejected
Scalar vs Residual	17.528	7	0.014	Rejected

The metric invariance model does not significantly worsen fit compared with the configural model, suggesting that the factor loadings are comparable across rookie and veteran players. However, scalar invariance is rejected, indicating non-equivalent intercepts across groups. The support for configural and metric invariance suggests that the latent structure operates similarly for rookie and veteran players. In practical terms, this means that offensive involvement and interior presence appear to be represented by the same indicators in both groups. However, because scalar invariance is not supported, direct comparisons of latent factor levels cannot be made with confidence. Thus, latent mean comparisons is not conducted.

5 Conclusion

The results suggest that NBA player performance is not adequately represented by a simple distinction between offensive and defensive ability. The revised SEM model provides

a substantially better representation of the observed data than the initial proposed model, by capturing player involvement and interior presence as latent constructs. Furthermore, offensive involvement is found to be a significant positive predictor of interior presence, indicating that players with greater offensive responsibility also tend to contribute more strongly to interior aspects of play.

The results further supports configural and metric invariance but not scalar invariance, suggesting that the latent structure is similar across groups while full measurement equivalence is not achieved.

Overall, the study demonstrates how SEM and CFA can be used to identify and evaluate latent performance dimensions in sports data, while also highlighting the importance of model refinement when theoretical assumptions are not supported by empirical evidence.

For future research, more player statistics could be used to improve the explanatory power of the factor constructs. For the rookie versus veteran part, a more balanced sample would also improve the reliability of the results.

Throughout the project we learned how SEM and CFA can be applied to real sports data. One challenge we met was that the original structure did not fit as well as expected and made us rethink the latent structure. This helped develop skills in model evaluation, interpretation, and connecting theory with empirical results.

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